Canadian Ocean Modelling Forum workshop: 29 May 2025
32 attendees online- 12 in person for discussions, and 17 in person for the panel discussion

1. **Ocean model development: Globally and in Canada - Paul Myers and Youyu Lu**

<u>Models used in Canada - All, please fill out!</u>

Group	Model	NEMO Version (Current and Future Goals)	Contact	Focus/Use
University of Alberta	NEMO + LIM/SI3 + BGC (online tracers) Offline: Ariane Ocean Parcels Configurations: https://canadian-nemo-ocean-modelling-forum-commuity-of-practice.readthedocs.io/en/latest/Institutions/UofA/Configurations/index.html	Current: 3.6 and 4.2.2 Testing: 5	Paul Myers Inge Deschepper Clark Pennelly Tahya Weiss-Gibbons	-Circulation/transport -Sea ice/ocean interaction -Intermediate and deep water formation -biogeochemical … -freshwater …

University of Manitoba	NEMO + LIM2 + BGC + ICB	3.4, 3.6	Juliana Marson Luiz Henrique da Silva Gabriela Wasielesky Madhurima Chakraborty Group portal: https://sites.google.com/view/portal-ocean/portal	- Circulation (Baffin Bay) -Iceberg melt plumes (release of freshwater in depth instead of at the surface) -Development of python package (ANHAlyze) for ANHA outputs -BGC in Hudson Bay and James Bay (AGRIF) - Testing longer simulation for iceberg season severity representation along east coast of Canada
ECCC-CCCma	NEMO + SI3 + BGC BGC: CMOC & CanOE =>https://gitlab.com/cccma/canesm (Earth system) =>https://gitlab.com/cccma/cannemo (ocean+ice+BGC)	Current:NEMO4.2 Testing: NEMO5	- Geoff Stanley - Nicolas Lambert (interim) - Natasha Ridenour - Damien Ringeisen (sea-ice) -	Coupled Climate modelization - Global eORCA1 & eORCA025 - Regional downscaling CanTODS+

University of BC	NEMO + BGC: SMELT + Carbon/Oxygen: SKOG and offline: WW3 (waves) Ariane Ocean Parcels MOHID Atlantis namelists: https://github.com/SalishSeaCast/SS-run-sets (current version v202111) code (private, ask for access): https://github.com/SalishSeaCast/NEMO-3.6-code main tools: https://github.com/SalishSeaCast/tools Main documentation (a bit dated) : https://salishsea-meopar-docs.readthedocs.io/en/latest/ our onboarding: https://ubc-moad-docs.readthedocs.io/en/latest/ and tools: https://salishsea-meopar-tools.readthedocs.io/en/latest/SalishSeaTools/index.html	NEMO 3.6 thinking of going to 5 but probably not this year Model Results (Hindcast + Realtime): https://salishsea.eos.ubc.ca/erddap/index.html	Susan Allen Doug Latornell	Salish Sea all things: circulation, mixing, Lagrangian advection, understanding nutrients, oxygen, carbon, ocean productivity, zooplankton. Downstream full ecosystem model including fishers. Going into the past and into the future. Pollution (oil, plastics, PBDEs etc)

DFO (Pacific)	FVCOM + BGC		Laura Bianucci	Process studies in the nearshore; Aquaculture; Future scenarios
DFO (NL)	NEMO + BGC	3.6 right now but will follow M. Dunphy's Port model version	Andry Ratsimandresy Thomas Guyondet	Process studies in the nearshore, aquaculture
DFO (Pacific)	NEMO + BGC: SMELT + Carbon/Oxygen: SKOG (SalishSeaCast from Susan Allen)	NEMO 3.6	Amber Holdsworth Michael Dunphy Natasha Ridenour	Climate downscaling for the Salish Sea
DFO (Pacific, Maritimes, Quebec)	NEMO	Currently 3.6, actively moving to 4.2 to enable use of Wet&Dry and AGRIF	Michael Dunphy Rachel Horwitz Stephanne Taylor Andy Lin Maxim Krassovski Simon St-Onge Drouin Hauke Blanken	Port models (Kitimat, Vancouver, Fraser River, Saint John Harbour, St Lawrence, Canso) Purpose: to support electronic navigation and emergency response

DFO (Pacific)	NEMO	3.6, with plans to move to 4.2 or 5 in the future; this will follow the updates to the ports	Michael Dunphy Andy Lin	RELIOPS -- RELocatable Ice-Ocean Prediction System. Currently we can downscale within CIOPS-W, adding CIOPS-E is work-in-progress. The aim is to be able to rapidly deploy a high-resolution downscale to support emergency response (spills, SAR, …) with automated model configuration (namelists,grid,inputs, etc) such that we only need to specify the region of interest.
DFO (Maritimes)	NEMO/LIM3	3.6	Youyu Lu, Xianmin Hu, Li Zhai	Ocean & sea-ice; Canada's Three Oceans domain (Pacific north of 44N & Atlantic north of 7N), ORCA grids, ¼ & 1/12-deg; Historical hindcasts 1958-present
Oregon State University	MOM6		Chengzhu (William) Xu	Tide modeling
ECCC (CMC)	NEMO3.6-4.2+CICE with coupling to WaveWatch3, GEM		Fred Dupont, Greg Smith, F. Roy, JF Lemieux, JP Paquin, Drew Peterson	Global, regional, coastal scales with coupling to atm and waves, no BCG on the horizon yet.
Dalhousie University	ROMS+CICE +WaveWatch3		Jinyu Sheng (Katja Fennel for BGC)	Coastal circulation, wave-current interactions (and BGC by Katja Fennel's group)

Past NEMO seminars:
https://canadian-nemo-ocean-modelling-forum-commuity-of-practice.readthedocs.io/en/latest/Seminar.html

SIGN UP FOR THE SEMINAR SERIES:
https://docs.google.com/forms/d/e/1FAIpQLSejoRsr-bj92uA1yKdWmb7neb4x7W8f9FmFb0zgqZLE2tw-TA/viewform?usp=sharing&ouid=114389673465925263102

NEMO Forum Website:
https://canadian-nemo-ocean-modelling-forum-commuity-of-practice.readthedocs.io/en/latest/

Overview of recent DFO oceanographic modelling: Han, G., Bianucci, L., Chassé, J., Foreman, M., Holden, J., Horwitz, R., Holdsworth, A., Lambert, N., Lavoie, D., Le Corre, N., Lu, Y., Peña, A., Ratsimandresy, A., Wang, Z., Perrie, W., Steiner, N., Thupaki, P., Zhai L., Brickman, D., Christian, J., Dunphy, M., Soontiens, N., Vagle, S., Guyondet, T., Page, P., Haigh, S., O'Flaherty-Sproul, M. 2022. Status of DFO Ocean Modelling: A National Review. Can. Tech. Rep. Hydrogr. Ocean Sci. 338: viii + 58 p.

- Fred Dupont-
    - There is a concern about using US-based models
    - The topic of concern is which language to continue. Oceananigans (Home · Oceananigans.jl) is a good example. How to attract qualified personnel with the use of Fortran as training is not in Fortran anymore at a uni base level
- David Greenberg
    - Using the Julia language seems to be an interesting one to use
    - SLIM in Belgium- May not be active anymore (JSterlin)
    - Suntans
    - SKISM- has sediment, BGC, and ice
    - TUGO or TUGOM—It takes in all the data as originally prescribed. For example, the model inputs and interpolates ERA5.
    - https://www.sciencedirect.com/science/article/abs/pii/S0278434307000374 Resolution issues in numerical models of oceanic and coastal circulation
- Fred Dupont
    - Sharing libraries

- Susan Allen
    - Main work development is in BGC
    - But physical modelling- surface waves using a wave model with parameterisation (Moore-Maley UBC thesis)
    - NEMO code base is on GitHub- If you want access, please ask sallen@eoas.ubc.ca
    - The Selish sea case mixing is too small- internal wave mixing scheme for 400m deep (writing up as part of the Oxygen paper)
    - Sewage outfalls (coming as part of the Alkalinity Enhancement paper)
    - Improving bathymetry (and trying double resolution)
    - Wetting and drying in future versions of NEMO- Who has done it with Z coordinates? I know I am not alone in thinking about: S. Taylor & M. Dunphy, for example.
- Youyu
    - Sharing is important, with consideration of credit for work previously done
- Yarisbel
    - On the mixing vertically- vertical resolution or parameterisation?
    - Susan's reply- The model has good resolution at the surface. Vertical resolution won't fix it, and NEMO was developed for the open ocean; it is difficult for closed or small seas.
    - Youyu reply- increase vertical resolution at a cost
- Nicolas Lambert
    - NEMO SI3, SMOC and CANOE- CanNEMO through CanESM (CanESM is the coupled model with ocean, atmosphere, land components)
    - Publicly available on GitLab https://gitlab.com/cccma
    - Using NEMO 4.2 but doing a big effort for NEMO5
    - Have expertise in the atmospheric and hydrological model coupling
    - Perspective for the future- The Iceberg model is activated, and ice cavities as well
    - We intend to not only share the model but also the infrastructure information (configuration and namelist) to be able to do downscaling (scripts for achieving it) by the public- Git feature
- Jim Christian
    - CanOE (NEMO base) and CMOC (not NEMO based and first used PICES as a structure as a whole, but it was not ideal)
        - Version shift is PISCES and may need to change the structure of your code so that changes to new versions are easier
- Version jump from 3.6 to 5
    - Nicolas- jump from 4.2 to 5 - Git merge function that made it easier. It is advised to go to the highest if

needed
    - Performance improvements- sold as 50% improvement, but not exactly 50%. More 30 to 35%
    - Michael- 3.6 to 4.2 is approx 30-40% speedup (Salish Sea, zps test)
- Fred Dupont
    - AGRIF (two-way nesting library) is a way in structured grid models to address different timestep requirements in different regions of the grid. [response to Dave Greenberg raising the utility of the adaptive timestep scheme implemented in Mog2D/TuGOM]
    - Which coordinates? Z or sigma coordinates
        - Heather replies- looking at vertical coordinates for users at the bottom. Would like to get references on the pressure gradient effects
        - David- z, partial step or chiselled, Greenberg et al. 2006 discuss resolution issues in numerical models. And Hannah and Wright 1995 (Depth-dependent analytical and numerical solutions for wind-driven flow in coastal ocean)
        - Problems with sigma over canyons: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2001JC000978
    - FABM BGC model- method to help get the BGC models into NEMO (GitHub - pmlmodelling/NEMO4.0-FABM).
    - AI use?- BGC and wave models are expensive to run. AI can emulate those components cheaply, and also provide mixing parameterisations to the ocean model.
    - UBC experimenting with SalishSeaCast ML emulator for spring bloom
    - Improve it or provide a simplified BGC version that would not be expensive
- Michael Dunphy
    - 4.2
    - Physics differences, for example, no Smagorinsky for tracers
    - Hybrid-coordinates to get wetting and drying in the port models- still in progress
- Natasha Ridenour
    - Forcing files: been working on a GitLab for processing boundary files and initial conditions for CanESM
    - Has worked on Glorys for CanTODS, there is a potential that this could be used for other configurations, but has not been tested
    - Also planning on running and testing with the JRA3 data, which is coming out soon
- Youyu
    - A package to prepare forcing data, most of the versions are in Python, from Michael Dunphy
- Natasha and Jonathan Izett

    - Current packages are in Python. New and not built from other scripts
    - It will be merged on GitHub, but it will take some time due to the CCCMA policies
- Youyu
    - Hindcasts- reanalysis fields some have less bias, and some have more. Unless you know the bias, you can't correct for it
    - Tuning albedo or parameters, if you know the forcing bias
    - Fred replied- interested in the methods used in the climate models. Basic AI is used to reduce the error. Potential to use it for the ECCC use for the climate forcings
    - AI for bias correction is an interesting tool: https://gmd.copernicus.org/preprints/gmd-2024-185/gmd-2024-185.pdf
    -
- Icebergs
    - Laura Gillard- who is using the iceberg model
    - Duo Yang asked about using it in Antarctica? A global run at UofA
    - Luiz- Has work been done on the geometry of icebergs?
    - Duo Yang- added the iceberg module in NEMO for Greenland and Antarctica. Seems to be interesting and has a remote impact on the climate
- Tahya
    - Small domains using the AGRIF nesting
    - Relocatable one-way nest or two-way nests
    - Susan- use a one-way nest for the Salish sea
    - Luiz -Paolo at UofM is looking into the Agrif nesting
    - Port models have multiple nests- 4.2 two-way with tides
    - Paul- Clark is playing with agrif tests in 4.2 and 5. But currently unstable
    - Fred- With tides, it is a bit stiff. One way is to modify NEMO with a … boundary: using a Flather boundary to get the tides to work, even one way (IF ANYONE HAS INFORMATION ON THIS PLEASE PASS IT ON TO SUSAN ALLEN AND MICHAEL DUNPHY)
    - Youyu- concern with baroclinic boundary layer: so need sponge as well
- Damien Ringeisen
    - SI3, BBM rheology
    - Who is interested in sea ice in Canada?
    - Youyu- ice wave interaction interest in marginal ice zone- ice fracturing changes? Size distribution in McGill
    - Inge- light importance for BGC

    - Paul- tidal mixing with sea ice issues and compare. How many categories is the best choice?
    - Fred- working on wave-ice interactions (Dany Dumont and Bruno Tremblay working on improving sea ice representation in the MIZ).
- Is there a white paper ideas from this discussion?
    - Paul- Make one of the main deliverables for the new 3-year funding application a white paper.
- NEMO-ICB References:
    - https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2021JC017542#
    - https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2023JC020697
    - https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2018GL077676

2. Biogeochemical models used in Canada and future development plans- Susan Allen and Inge Deschepper

BGC table - All, please fill out!

Group	Model	What are you working on	What are you stuck on	Contact
University of BC	SMELT https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2019JC015766 Carbon (SKOG) https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2021GB007024 Oxygen in draft form	Oxygen Zooplankton Carbon impact on plankton Outfall/Alkalinity Enhancement Bacteria Turbidity	Sediment impact wo a sediment model Finding good studies that constrain at least two carbon system parameters	A. Tall K. Suchy K. Suchy D. Ianson S. Allen C. Stang A. Tall J. Valenti E. Olson S. Allen

		River input		
University of Alberta	BiGCIIM and BLING	Sympagic Carbon Upgrade from 3.6 to 4.2.2 or 5 Nutrient transports	Numerical instability due to tides in Foxe Basin Sediment burial fraction	Paul Myers Inge Deschepper
DFO (NL/Maritime)	CanOE	Mussel carrying capacity. Bivalve ecophysiology	For now, tuning the physical model	Thomas Gyuondet Andry Ratsimandresy
DFO/CCCMa	CanOE	tuning community composition for phytoplankton/zooplankton groups	We added a simple sediment model, but it didn't make much difference for our 2km model of the BC coast - we wonder if it may be more impactful in higher resolution domains.	Holdsworth Christian
ECCC-CCCma	CanOE, CMOC	Global modelling (CanESM), Regional modelling/downscaling (CanTODS)		Jim Christian Krysten Rutherford Elise Olson
Dalhousie	bio_Fennel (ROMS)	Dalhousie's Marine Environmental Modelling Group		Katja Fennel
DFO (Pacific)	FVCOM-ICM	Fjords modelling (oxygen, submerged aquatic vegetation)		Laura Bianucci

DFO (Pacific)	ROMS-BGCM			Angelica Pena
DFO (Pacific)	CanOE, CMOC	Hindcasts and projections of Canadian Arctic; sea-ice bgc		Nadja Steiner

- What are people developing: river nutrient sources, oxygen, carbon, ice/snow, light attenuation
- Stuck on: light attenuation/turbidity, river nutrient sources, carbon, ice/snow, detritus, sediments, bacteria, oxygen

1. Ice/snow

- Snow fields will be crucial for light processes, heat flux, etc

Jim
- Developing sea ice BGC, previously done in an older version of LIM, Antoine is now incorporating sea ice BGC into NEMO4.2 with SI3 in CanESM
- Hopefully, it will be available soon to other groups

Inge
- BiGCIIM does have a sympagic component. Inge is working on upgrading NEMO and joining BiGCIIM
- With the new ice module, it will be more complicated
- Light attenuation is an important topic, for pelagic and sympagic modules, are we actually representing under ice blooms, because you only see chl if you have an open water fraction (joint project from CAMAS)
- It would be great to do some experiments/testing on this from a physics perspective
- Natasha: Elise and Krysten talked about CanOE and there were some issues with PAR in the CanTODS domain. Can be contact points for this
- Jim: New sea ice modules do transmission of light through ice and snow differently than old ones did. BGC modules developed with LIM2 assumed irradiation transmittance was done a certain way, and it isn't the case
- How well is snow on the ice handled in the new models? LIM2 is not handled; if there's any snow, it blocks light
- As the models become more complex, we have to make sure light attenuation is correctly handled

Paul
- Question of resolution in models: If we have high-resolution physical models and are looking at potentially

coarsening the physical fields to drive the BGC model offline, are there studies that look at these processes?
- Inge: Mesoscale eddies show PP blooms being altered by physical processes
- Jim: CanTODS and ¼ degree: certain physical processes that drive BGC distributions are probably not resolved at this resolution
- Susan: How do we know when we have the best resolution for accurately describing the physical properties?
- It’s hard to quantify if the model is correctly estimating BGC distributions because of a lack of data; we probably need to run at a 12th degree or higher to start to resolve these

2. Light
    a. Attenuation
    b. Turbidity
3. Base nutrients
    a. Macronutrients
    b. Micronutrients
4. Freshwater sources
    a. River contributions

Tahya:
- Involved with a group through CAMAS on the comparison of the McKenzie River delta to look at different models
- Getting a good nutrient input in the Mackenzie in this area, then other rivers
- Data from the Great Rivers Arctic Observatory (nutrients)
- Potentially getting nutrient data from the HYPE hydrological model itself

Natasha:
- Doing downscaling for river discharge, looking for the land group to add chemistry and temperature for rivers
- Land surface model part of CanESM, issues from remapping rivers to higher resolution, going from 1 degree to 1/12 degree
- Constituents: nitrogen, silicate
    - Should have 2 of the carbon cycle: pCO2, pH, ALK, DIC
    - CDOM would be good to add, links to light attenuation and turbidity along the coastlines
    - Any form of dissolved or particulate organic matter
    - Temp: We should have temperature data coming from A-HYPE

Paul:

- We should add nutrients from ice sheets
- Glacial runoff + icebergs
- Is there anyone with this kind of data, or who knows about existing datasets?
- Oxygen isotopes, Methane and helium are good markers which can be used partially as tracers
- Obs can be sparse.
- Laura - If we had a local observation test region - how could we implement this in the biogeochemical model - or which model would be best for looking at oxygen isotopes, methanes, helium?

5. Zooplankton
6. Bacteria
    a. Implicit vs explicit
7. Detritus/sediments

Jim/Amber
- CanOE has a new sediment model that goes with NEP36 that Amber developed (NEMO3.6)
    - "We added a simple sediment model, but it didn't make much difference for our 2km model of the BC coast - we wonder if it may be more impactful in higher resolution domains."(from table)
    - Had an effect on correcting DIC a little bit, but had sparse obs anyway
    - Could it be tweaked slightly to get a better result? Maybe, but it was a lot of effort for small payoff

Inge
- BLING no sediment at bottom, it's remineralized before
- BiGCIIM - trap component and added a burial fraction but that was more for the control of mass/balance, than looking into sediment, there's no oxygen in this model so we can't look at that now
    - Is there a difference in these models?
        - They do act slightly differently - the burial component controls drift a bit better, but BLING doesn't have as much drift to start with so hard to tell
        - Issues in very shallow waters (15-20m depth) with things exploding, Foxe basin bathymetry is poorly known at this stage.
- Burial: in CanOE there's burial of calcite carbon, don't have that in CMOC (idea is if it gets to the floor it gets buried), whereas in CanOE, there's burial or dissolution depending on saturation state, this becomes important in shallow waters where saturation state can be high
    - Calculate saturation state, which is greater than 1, 100% burial, if it's less than 1, 100% dissolution
    - Can be expensive bc you have to solve carbon chemistry at all depths, vs in CMOC it's only solved at the

surface.

Susan

- Oxygen variations in models funded by Seattle area, how much diatoms sink vs how much are remineralized, and PON, how much is remineralised vs buried, resuspension of sediments in salish sea is common, so part of the sediment that reaches the bottom is reflected
- Struggling with getting oxygen right in terminal bays in Puget Sound, timing is difficult to get right, also have bathymetry problems south of the border, so that's also an issue we don't need to deal with in Canadian waters

Inge: Sedimentation of the matter: in Hudson Bay, there's high river water inflowing that is taken into account for sedimentation speeds, and we use a salinity relationship.

- For detritus sinking speed, Salish Sea Cast uses a constant and looks at how nutrients look with depth to see if the speed is right
- CanOE: sinking rates are constant for each class, 2 m/day for small, ? m/day for large
- Inge: Salinity dependent sinking rate: from 0-15 salinity, higher sinking speed, from 15-30: slightly slower, higher than 35, slowest sinking speeds, to try and accommodate for particulate organic matter from rivers
- Lots of models are developed for the open ocean, but sinking ballast can be dependent on the open ocean/coastal
    - https://www.sciencedirect.com/science/article/pii/096706379390077G

8. Carbon

- What do we need to know about carbon to model it well
    - Inorganic and organic
    - The main quantities of the inorganic carbon system have two main parameters for modellers, total alkalinity and DIC, which have to be linked with biology bc PP are taking up DIC
    - Also, need detritus (in whatever your variable is) can then convert that to carbon (Redfield ratio)
- Anthropogenic carbon
    - Susan - currently working on making boundary conditions for pre-industrial run, using apparent oxygen utilisation, figure out an estimate of pre-industrial carbon, then can estimate CO2 in the atmosphere at that time
    - David
        - Stu Smith and Peter Jones paper on carbon
        - Could account for all carbon you needed to extract from the atmosphere through down(?)welling front
        - Recently, DAL is hiring carbon chemistry: cannot observe carbon intake/uptake at a model scale
    - DFO has good observations for total alkalinity and DIC, and the Salish Sea model matches well (Susan)

    - Salish Sea doesn't have upwelling/downwelling so this could change the ability of the model to capture this process properly

Papers:

Smith, Stuart D., and E. Peter Jones. "Isotopic and micrometeorological ocean CO2 fluxes: Different time and space scales." Journal of Geophysical Research: Oceans 91.C9 (1986): 10529-10532.

https://www.sciencedirect.com/science/article/abs/pii/096706379390077G (Does "Novara Knoll" exist?, Deep Sea Research Part I: Oceanographic Research Papers)

3. **Cross-institute coordination of NEMO development- Frederic Dupont and Natasha Ridenour**

Joining the NEMO consortium will require one full-time employee.

- Contributions that Canada could add to the NEMO code are listed in the slides from Fred, but could include AGRIF development, special SSH BBC, wave-coupling interface etc.
    - AGRIF nests with polygons instead of regular rectangles are being tested at NOC (Adam Blaker?)
    - David: Using irregular AGIRF domains could be advantageous, especially ifyou can run multiple AGIRF domains in the parent at different resolutions
    - Paul: AGRIF often has a much smaller overhead to run, the outer domains take much less time to run, with most of the resources to run the inner child
    - Clark: In 3.6, the processors didn't run in parallel for the parent and the child. This has likely not changed in 4.2, though this condition likely has been relaxed. In 5, should be able to run in parallel, though no one has tested this yet, it seems
        - Parallel sisters are available in NEMO 5

Slack channel invitation that is owned and run by Susan Allen:
https://join.slack.com/t/nemo-canada/shared_invite/zt-36eslc7j7-EeDa_SLkuBmpbqC8mC0n3g

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LIM3 vs SI3 vs CICE

- Fred: LIM3 has been shown to have good performance, will need to show that we can get similar results with SI3
- CICE is likely more advanced, but being part of the NEMO consortium will probably require a full switch to SI3. Though it could stay involved with the CICE consortium and include the updates in the SI3 module

Icebergs
- While some of Juliana's iceberg developments have been pushed to the NEMO code base, there are still ongoing developments for this in Canada (melt plume parametrisations etc)
- There are also still desired iceberg developments, such as icebergs in AGRIF. Currently, icebergs are only allowed within a nest and cannot pass across a boundary

NEMO 4 vs NEMO 5
- Paul: What version should we be using for coordination? Will the government be sticking with NEMO4?
- Fred: Any contributions we want to give to the NEMO consortium will have to be through NEMO 5
- Nicolas: In the early stages of working with NEMO5, there doesn't seem to be big changes in the results between the two versions. There are changes in the speed which are most noticeable. More namelist options in NEMO 5. If you develop new code, you will have to pay attention to the size of your array. It was a Git merge to do it.
- It is normal that operational forecasting centres run older versions in operations, so don't mind us if we stick to NEMO4!
- Clark: NEMO 5 tests have been slow because of computers, but there are some issues with tides. We are trying to get operational at the moment.
- Fred: Is there pressure to transition from CPU to GPU? DRAC is more driven by CPU for us than GPU.
- Duo: I'd just like to comment that incorporation of a new NEMO version technically might be much quicker than tuning. We've spent a lot of time to test various parameterization schemes and parameters in NEMO4, including v4.0.3 and 4.2.
- Links from Jim: https://cdn-assets.inwink.com/e2ed7ccb-dbb5-4dc9-9b30-9a6876d1fc19/aa24209b-6893-41eb-af45-abb14afad593
- https://github.com/pmlmodelling/NEMO4.0-FABM

Canada's involvement in the NEMO consortium
- CoP funding for Full-time employee?
- It should be the government to represent Canada, but if it would be easier, it could be through academics
- Fred- CCCMa's involvement, community-wise? Issue with being inside the firewall.
- Susan- starting to see issues with firewalls in academia, but not as strict yet. Strongest coders in the government,

so it is important
- Fred- NEMO officer needs to be a 0.5 FET and then 2 x 0.25 FETs - Where can we get the funding on the long term?
- Youyu- managers at DFO are very reluctant to hire a permanent employee. Can be a research associate position in academia. Xiamen Hu is a strong candidate.
- Heather- it is being brought up by CONCEPTS to the directors at DFO.
- Paul- are there more academic links in CONCEPTS that can be done?
- Heather- Need to get a sense of where the funding is already being spread, as it is very regional on the DFO base. CSA is no longer part of the CONCEPTS, and involvement in that form is needed.

Current barriers that we can try to remove to help with coordination?
- Nemo Seminars
- Online forum- Susan's Slack
    - Nicolas- the government firewall blocks Slack
- Sharing methods (Git)
- Webpage updates
- Shared vision?
    - Testing opportunities to work in different groups
    - Use funding for internships
    - Fred- Is a programming language blocking people from coming into ocean modelling?
        - Tahya not really
        - Susan took with them the coding languages and version control at the end of the degree
        - Inge, not really, because I had to learn it no matter what
    - Laura- could students get credits with collaboration with CCCMa or DFO, a co-op opportunity?
        - Susan- Coop isn't a credit but it is registered on transcripts. But our degrees are long so adding the coop would extend the time.
        - Natasha- Could there be a way to align their project with the government co-ops closely?
        -
- Mailing list
    - Do we do a newsletter, or do we forward what people pass on?

        -Susan- The newsletter is a lot of work and may not be read

        -David- Canadian Ocean researcher, newsletter editor. He can pass it on to him (https://cncscor.ca/site/scor?language=en_CA) (sign up here:

https://www.mailman.srv.ualberta.ca/mailman/listinfo/cnc-scor)
- Natasha votes for informal- less work, and people will read it if it is simple
- Heather- what work is needed and inputted? What kind of interactions are people aiming to do in the group?
- Paul- How frequently do we want to do the workshops? Hybrid at least
- Natasha- Government attendance is limited
-Paul- we have funding to use to send ECRs for collaborations and hackathons

- Common tools
    - We could merge the tools into different groups
    - David- LLNL tools. Pocview (CNRS tool- https://github.com/jepler/poc)
    - Tahya and Luiz- ANHALYZE tools (https://github.com/PORTAL-CEOS/ANHALYZE)
    - Paul- ncview
    - Susan- added their link to the table at the top

4. **ERC panel: Careers and pathways in modelling in Canada**

5. **Wrap-up**

What activities should we consider going forward
- Which group is working on what? Continue with this type of meeting.
- White paper
    - Inter-model comparison
    - 5 pages, doesn't need to be long
- This type of workshop is the reason we come to conferences in person like this
    - The NEMO community has some momentum at this point
- We have a great working table of models we are all working on, it would be great to put this on the website and keep it updated as a resource of what's happening in Canada
    - This table could be the start of the white paper
- These types of workshops are great for ECR because it provides a blueprint of how people interact and work together

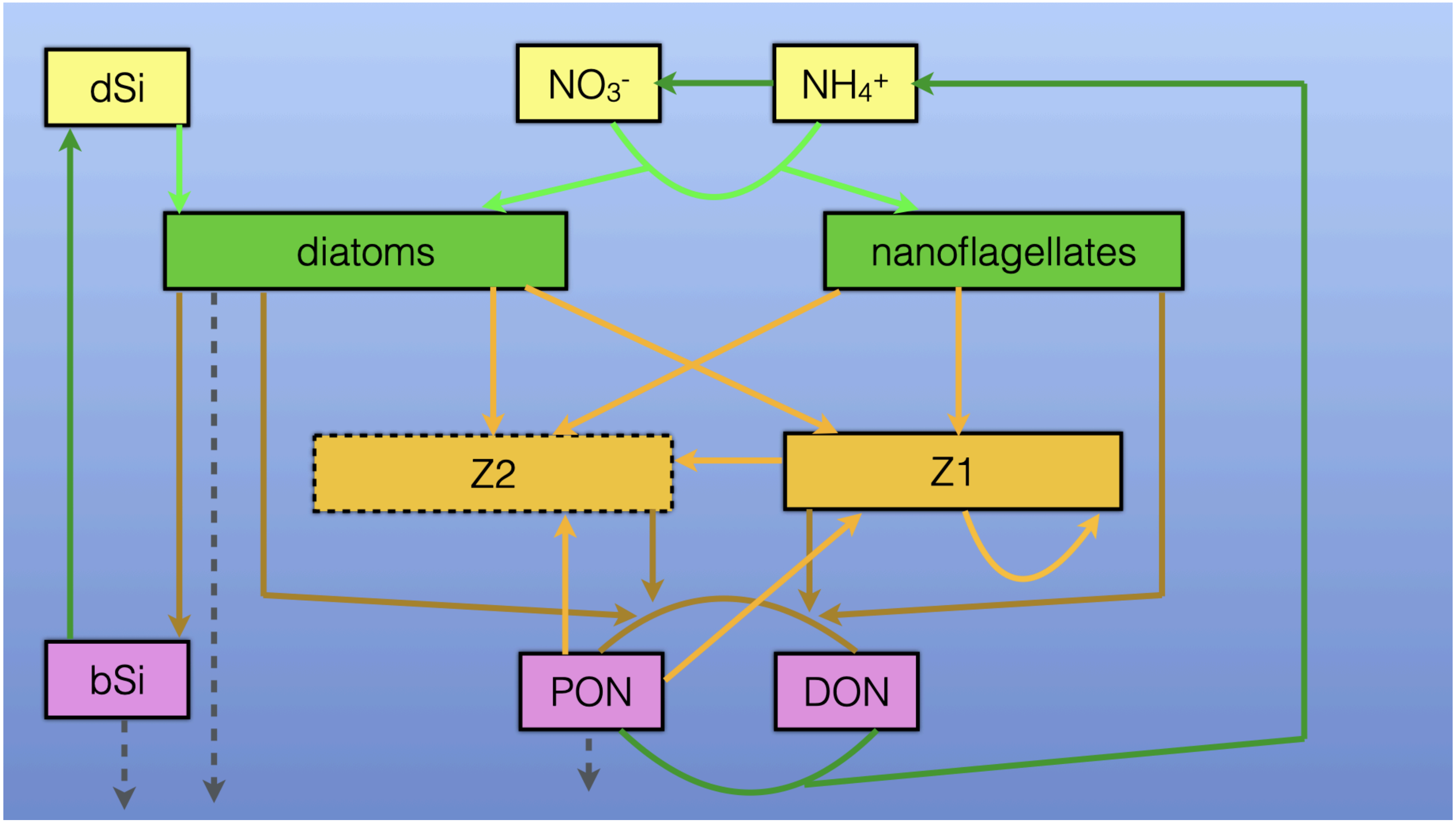
    - You can learn things you wouldn't otherwise know (how to apply for funding, how to write proposals, etc)
- Some government groups hire from students with modelling experience, DFO, ECCC, etc
    - The Canadian modelling community is not so big that having students in the community as they learn modelling will help the community grow and increase the quality of modelling in Canada
    - NEMO consortium can help with bug fixing, feedback on the model, and this is all beneficial, and we're contributing to Canada and the international community
- A couple of years ago, Guoqi Han published a DFO report of many used models. This could be a great idea to try and replicate or update (Can Heather add the link to this paper here?)
    - National review (2022): https://publications.gc.ca/collections/collection_2022/mpo-dfo/Fs97-18-338-eng.pdf
    - BGC specific review (2025): https://publications.gc.ca/collections/collection_2025/mpo-dfo/Fs97-18-388-eng.pdf

- Do we want to include other models going forward? Future discussion on this .

6. Important chat messages

Meeting chat is uploaded on the website and can be found here: https://canadian-nemo-ocean-modelling-forum-commuity-of-practice.readthedocs.io/en/latest/

Smelt (v202111)

7. 

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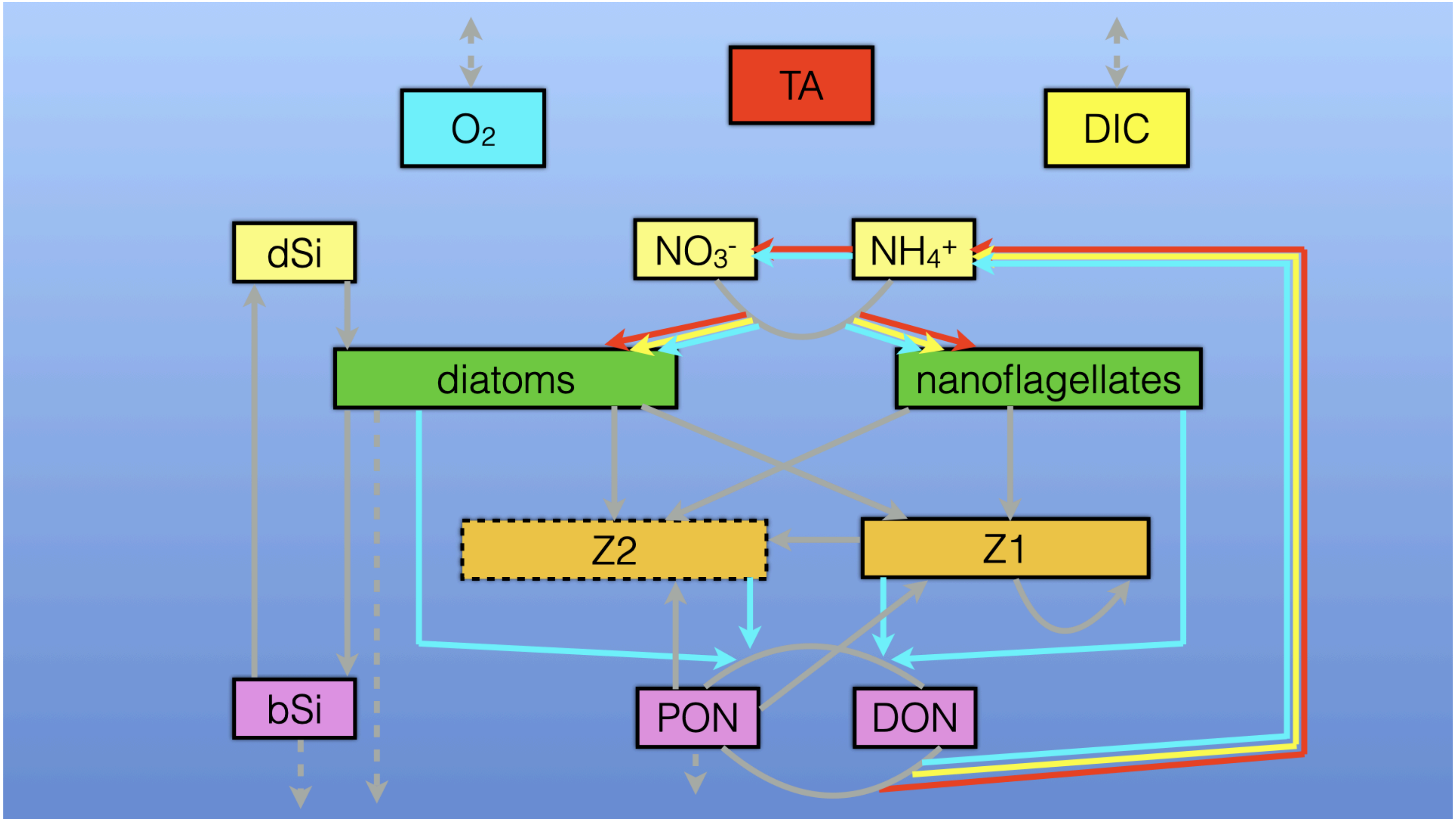

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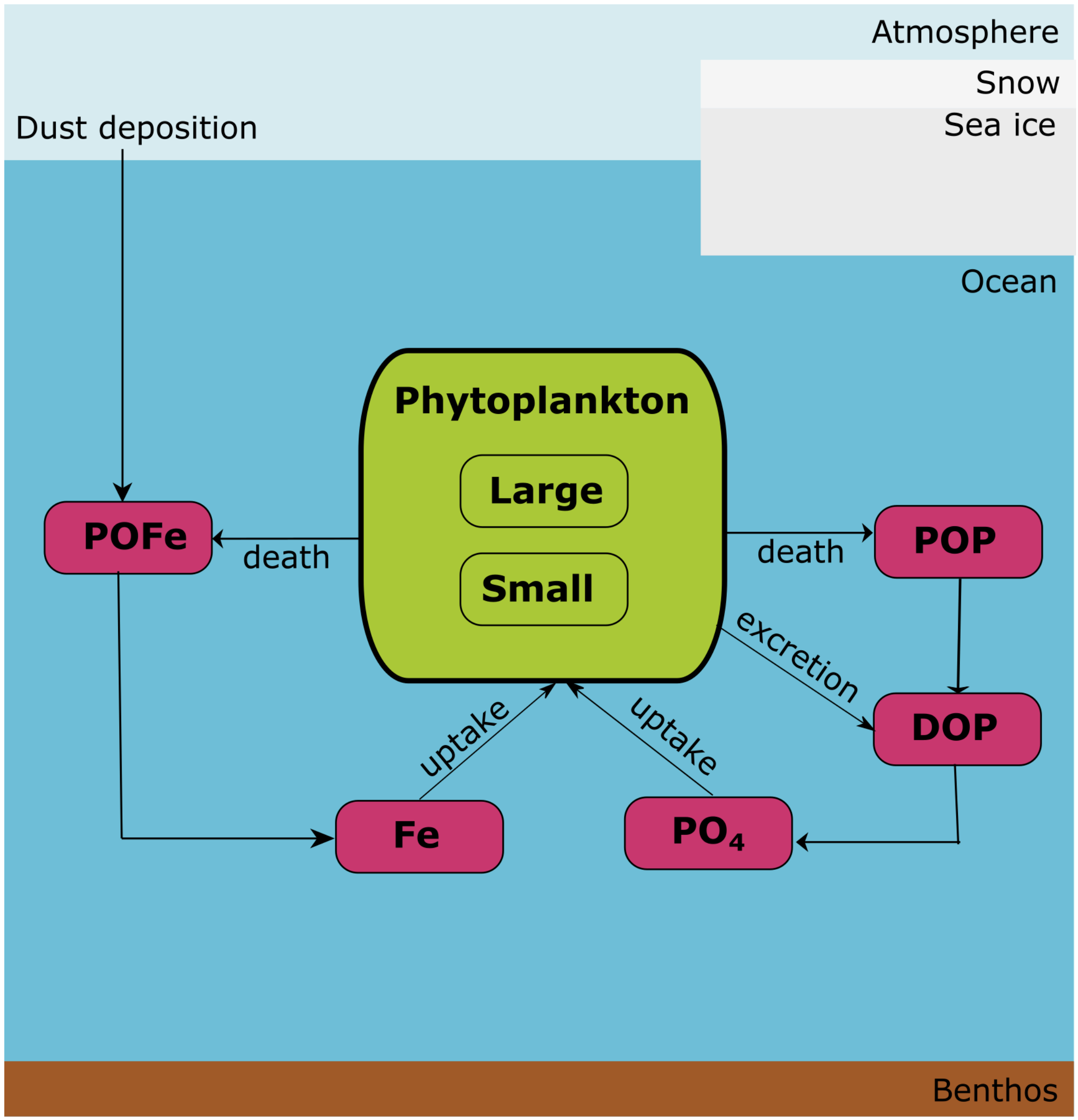
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Dust deposition
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Benthos

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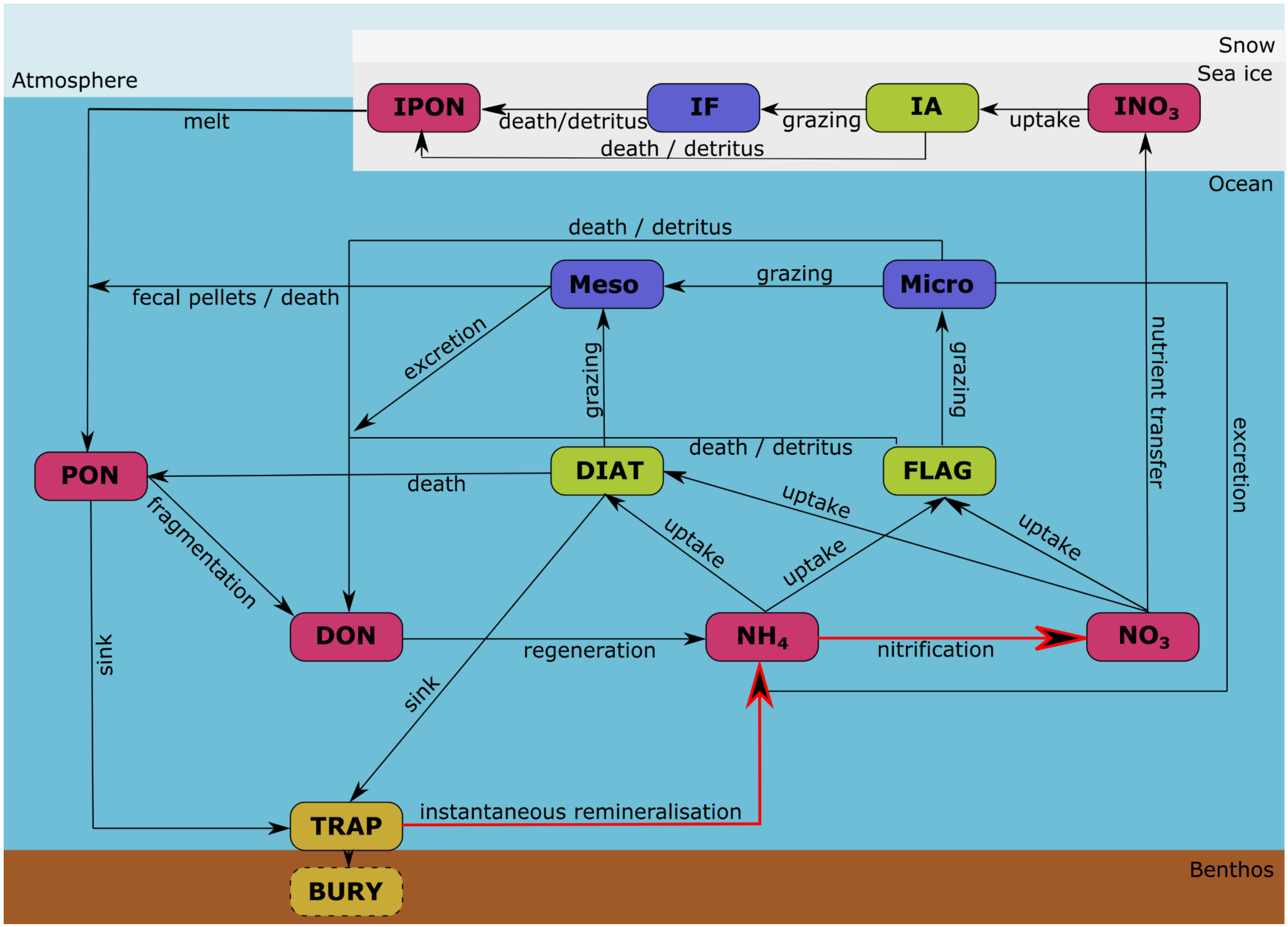

Snow
Sea ice
Atmosphere
Ocean
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nutrient transfer
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PON
DIAT
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death
fragmentation
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uptake
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uptake
DON
NH_4
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regeneration
nitrification
sink
sink
TRAP
instantaneous remineralisation
BURY